

DQ AoA2 fndEWatching Neural Networks Learn

N Nisanon-linear mathematical toolintendedtosimulatethehumanbrainprocessbyusingsimpleunitscalledartificialneurons.

N Nisauniversal function approximator.Theyaremachines thatcreate functions.

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N Nwithin finiteneurons can approximate any function given the right weights. How to achieve this or how to come closer should be the goal.

N Ncan be represented using matrices and vectors, where each neuron's input and output can be described as a linear combination of its inputs.

N Ncan be represented as directed graphs, where neurons correspond to nodes and connections between neurons correspond to edges.

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N Nare trained by optimizing a loss function with respect to the network's parameters. Optimization theory provides a framework for this.

N Nthrough techniques like gradient descent and its variants. Understanding derivatives is crucial for computing gradients, which are used to update the weights.

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N N. For example the former can be used to preparing or preprocessing of data, model evaluation and bringing into context the results.

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N Ncan be viewed as function approximators, and functional analysis provides tools for analyzing the properties of these approximators.

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Mathematical Fields	Formalization	Data Processing Model	Architecture	Optimization Parameter	Performance Evaluation	Generalization Measure
Linear Algebra Graph Theory Information Theory Optimization (e.g., Calculus) Probability Theory (Stochastic) Functional Analysis						

*NN, for example :
convolutional, recursive. This segment will focus on feed-forward networks. A Feed-
Forward is usually a non-
linear regression or classification with a sigmoidal activation and a multilayer perceptron.*

A ML/Sc035 Types of artificial neural networks.